

R-Value: Pricing NBA Production Against True Team Expenditure

Introduction. Public player-value metrics evaluate production but fail to take the associated costs into account. Nominal salary greatly underestimates those costs: under the luxury tax, a dollar paid by a repeater taxpayer can cost the franchise several times one paid by a team below the tax line. We ask which players and teams achieve the highest production per dollar of true expenditure, and whether cost-efficiency is connected to winning championships. The question is timely, since the apron system introduced in the 2023 CBA was specifically intended to curb high-spending rosters.

Method. R-Value matches a performance core with a CBA-correct cost model. The core combines five z-scored components z_k —box production (rate and volume), usage-scaled shooting efficiency, minute-shrunk on-court offensive and defensive impact, and PIE—into a composite index, applies the normal cumulative distribution function Φ to it, and adjusts for availability over m minutes:

$$\text{CPI} = \sum_{k=1}^5 w_k z_k, \quad \text{PS} = 100 \Phi(\text{CPI}), \quad \text{WPS} = \text{PS}(0.55 + 0.45A), \quad A = \sqrt{\frac{\min(m, 2200)}{2200}}.$$

The weights w_k come from a two-stage simplex grid search maximising All-NBA recall@15 and team-win rank correlation. The cost model computes each season’s luxury tax $T(P)$ on payroll P from the bracket schedule then in effect (2017 CBA through 2022–23, the 2023 CBA thereafter), applies repeater rates, and grosses up each salary s_i by the team’s total-expenditure multiplier:

$$\text{TE} = P + T(P), \quad \text{CI}_i = \frac{s_i (\text{TE}/P)}{\text{TE}}, \quad R_i = 100 \cdot \text{norm} \left[\text{WPS}_i \left(\frac{\overline{\text{CI}}}{\text{CI}_i} \right)^{0.35} \right],$$

normalised so the league average is 100. Team R-Value divides minute-weighted output by total expenditure. Data covers six seasons—2018–19 and 2021–22 through 2025–26—excluding the COVID-disrupted 2019–20 and 2020–21, whose irregular schedules and pro-rated tax settlements undermine the availability and cost models respectively.

Results. Under leave-one-season-out cross-validation—each parameter refitted while the season being scored is excluded—the core recovers 78.9% of All-NBA selections in its top 15, against 71.1% for PIE×minutes, 34.4% for raw plus-minus and 23.3% for estimated net rating, and correlates with team wins at $\rho = 0.914$ (Table 1). Weights are stable: four of six folds select the same vector, the offensive-impact weight identical in all six.

Championships are generally not obtained efficiently (Figure 1). Among the six champions, production-per-dollar ranked 16th, 29th, 14th, 13th, 3rd and 14th of 30. The 2021–22 Warriors won at an R-Value of 49, on a \$179M payroll carrying a \$191M repeater tax. A fitted title score weights eight-man rotation quality roughly nine times the top-two star premium, and ranks actual champions 2.83rd of 30 out of sample. Across 2025–26’s 582 players, R-Value is positively skewed at +1.04 (median 94.9 against a mean fixed at 100): most contracts yield below-average value per dollar, while a few rookie-level agreements return multiples of it.

Conclusion. The rosters offering the best cost value are seldom the ones that go on to win titles; instead, teams achieve competitiveness by spending through these short value windows. R-Value captures that trade-off in CBA-accurate dollars, letting front offices assess roster decisions on actual expenditure rather than nominal salary, and ranks players, evaluates teams and forecasts seasons from one model.

Code and data: <https://github.com/Shreyas23x/r-value>

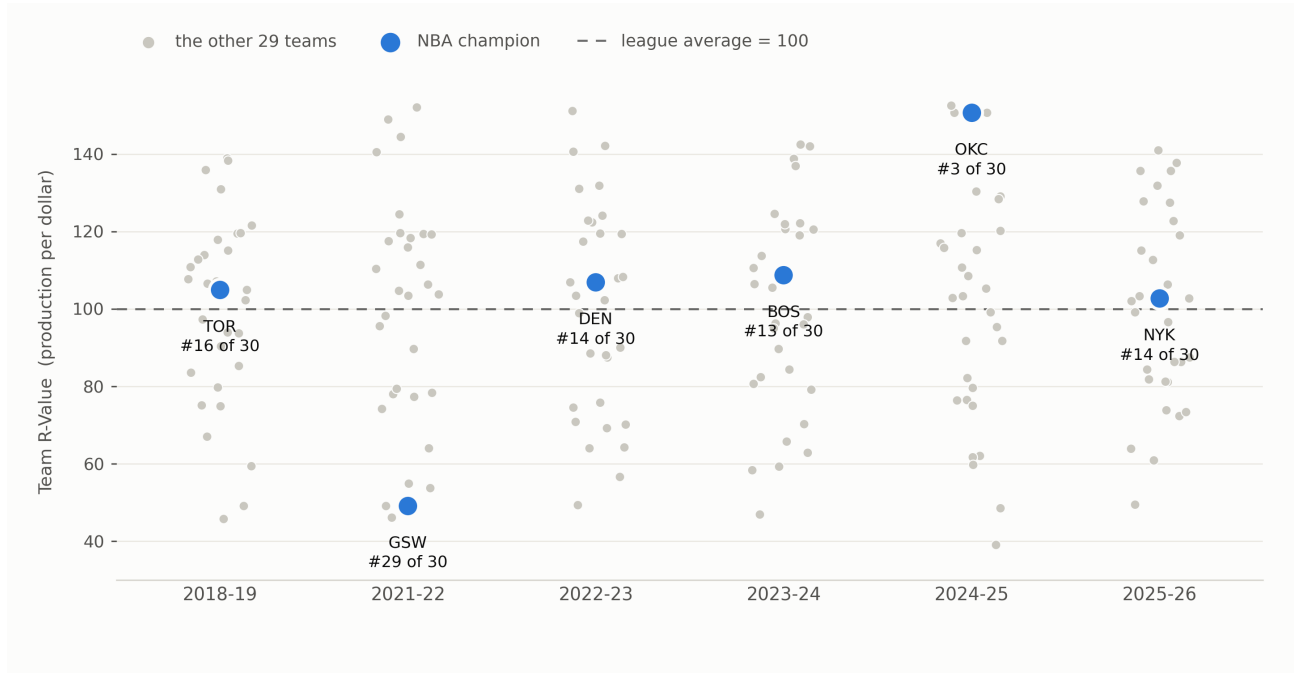


Figure 1: The R-Value for all 30 teams over six seasons, with the champion indicated. Five of the six champions are at or below the league average of 100.

Table 1: The leave-one-season-out cross-validation method, applied by refitting each parameter—the five pillar weights and the two title premiums—on the other five seasons and then evaluating it against the season left out.

Held-out season	R-Value recall@15	ρ with wins	Champion title rank
2018–19	0.800	0.935	1
2021–22	0.800	0.882	3
2022–23	0.733	0.880	6
2023–24	0.800	0.926	1
2024–25	0.800	0.932	3
2025–26	0.800	0.927	3
Average (out-of-sample)	0.789	0.914	2.83 of 30

Benchmarks with no fitted parameters: PIE×minutes 0.711, raw plus-minus 0.344, estimated net rating 0.233.